

Uniform convergence of Pfaffian point process to the Airy line ensemble

Zhengye Zhou

Abstract

We consider a family of Pfaffian Schur processes whose first coordinate marginal relates to the half-space geometric last passage percolation. We show that the line ensembles corresponding to the Pfaffian Schur processes with geometric weights converge uniformly over compact sets to the Airy line ensemble. By detailed asymptotic analysis of the kernels, we can verify the conditions for the finite dimensional weak convergence introduced in [7]. By utilizing the tightness criteria of the line ensembles established in [8], we can further improve the finite dimensional convergence to the uniform convergence over compact sets. Moreover, using the same methodology we also show that sequences of spiked Pfaffian Schur processes converge uniformly over compact sets to the Airy wanderer line ensembles constructed in [7].

Contents

1	Introduction and main results	1
2	Definitions	3
2.1	Airy line ensemble	3
2.2	Pfaffian Schur process	4
2.3	Correlation kernel and deformation of contours	6
3	Kernel Convergence	8
3.1	Asymptotic analysis	8
3.2	Limits of the kernel	12
4	Upper tail estimates	18
4.1	Kernel estimate	18
4.2	Probability estimate	20
5	Weak convergence of line ensembles	20
5.1	Finite dimensional convergence	21
5.2	Tightness of the line ensemble	23

1 Introduction and main results

In recent years, models related to KPZ growth in a half-space were extensively studied. For example, [1] studied the fluctuations of last passage percolation with exponential weights in a half-quadrant within the framework of Pfaffian Schur processes, [17] shows the fluctuations of a one-dimensional polynuclear growth model in a half-space, the log-gamma polymer in a half space was also considered [13]. For a large amount of half-space models, the fluctuations follow the GUE Tracy Widom distribution in the bulk. It is interesting to further investigate if these models converges to the Airy line ensemble.

The Airy line ensemble $\mathcal{L}^{\text{Airy}}$ is a random sequence of continuous functions that arises as a scaling limit in random matrix theory and various models within the KPZ universality class [5, 6, 15, 18]. In particular,